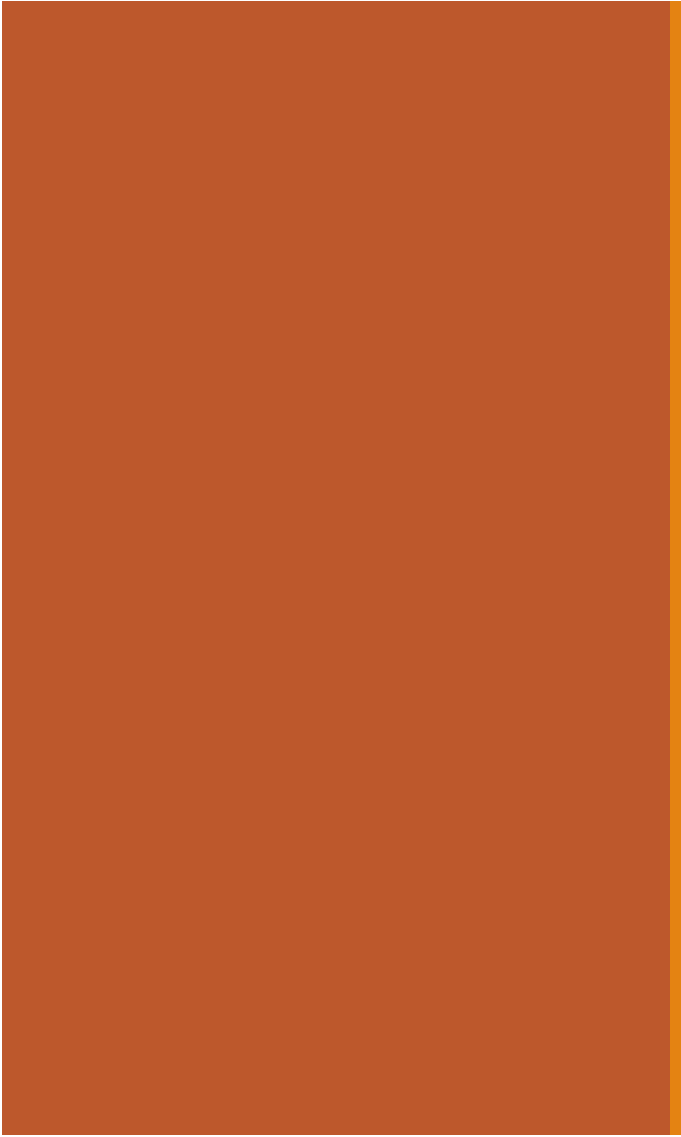


23: Naïve Bayes

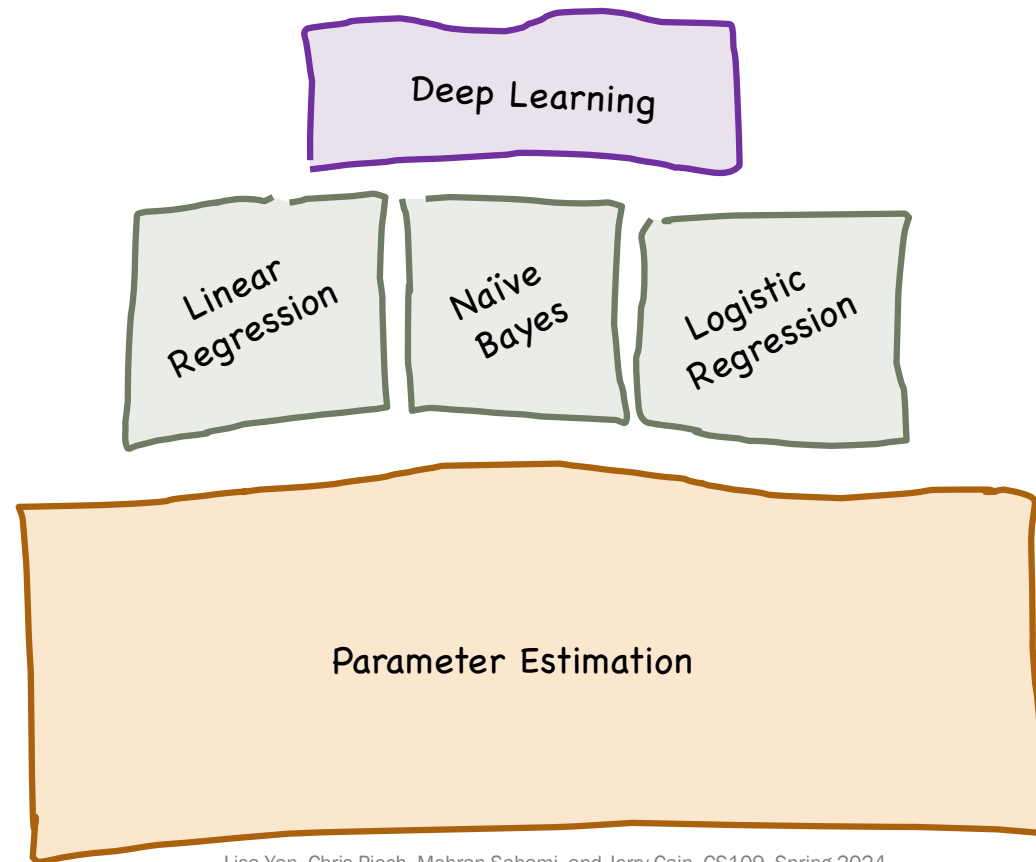
Jerry Cain
May 22, 2024

[Lecture Discussion on Ed](#)



Preamble: Machine Learning

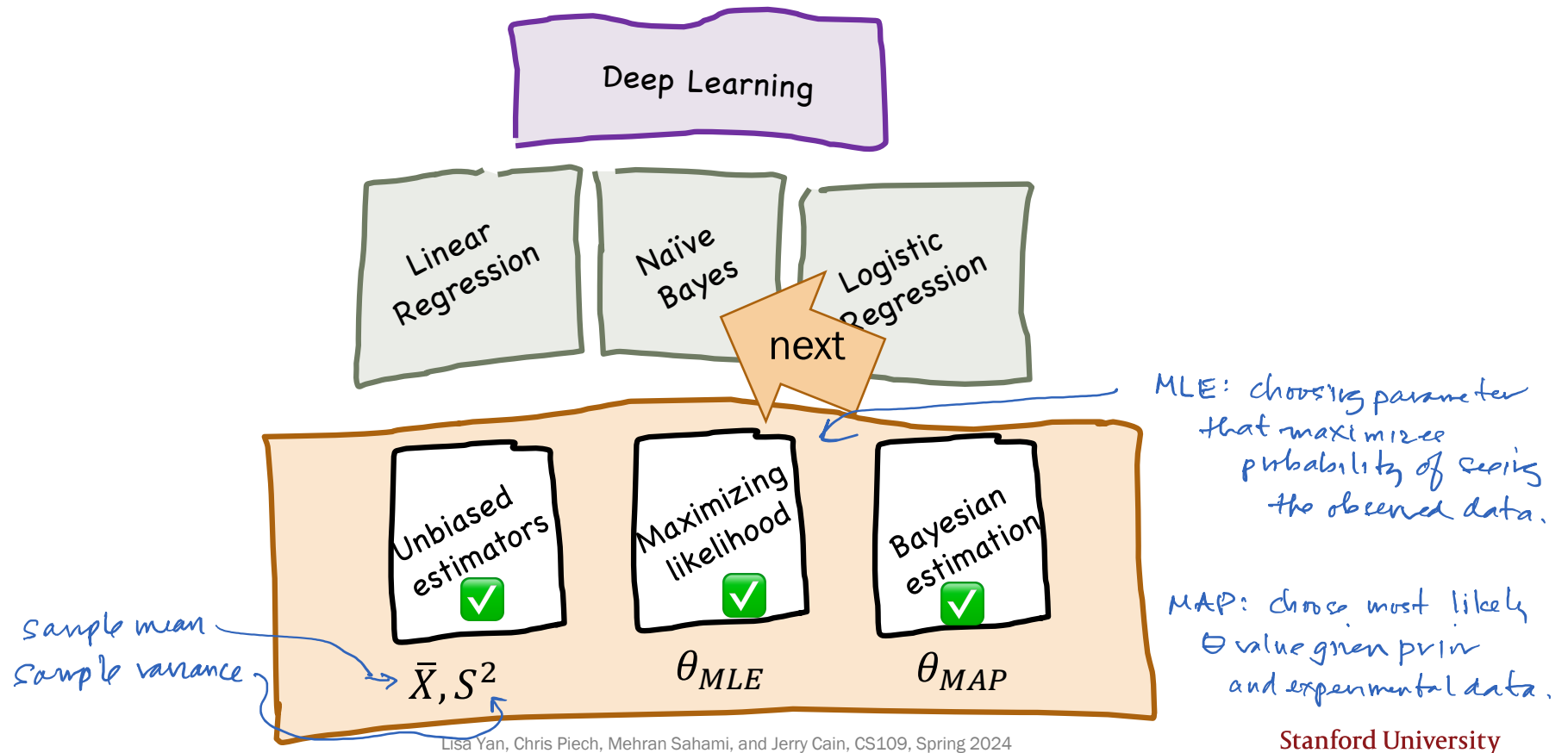
The Path Before Us



Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024

Stanford University

The Path Before Us



Machine Learning uses a lot of data.

Many different forms of machine learning

- We focus on the problem of **prediction** given prior observations.

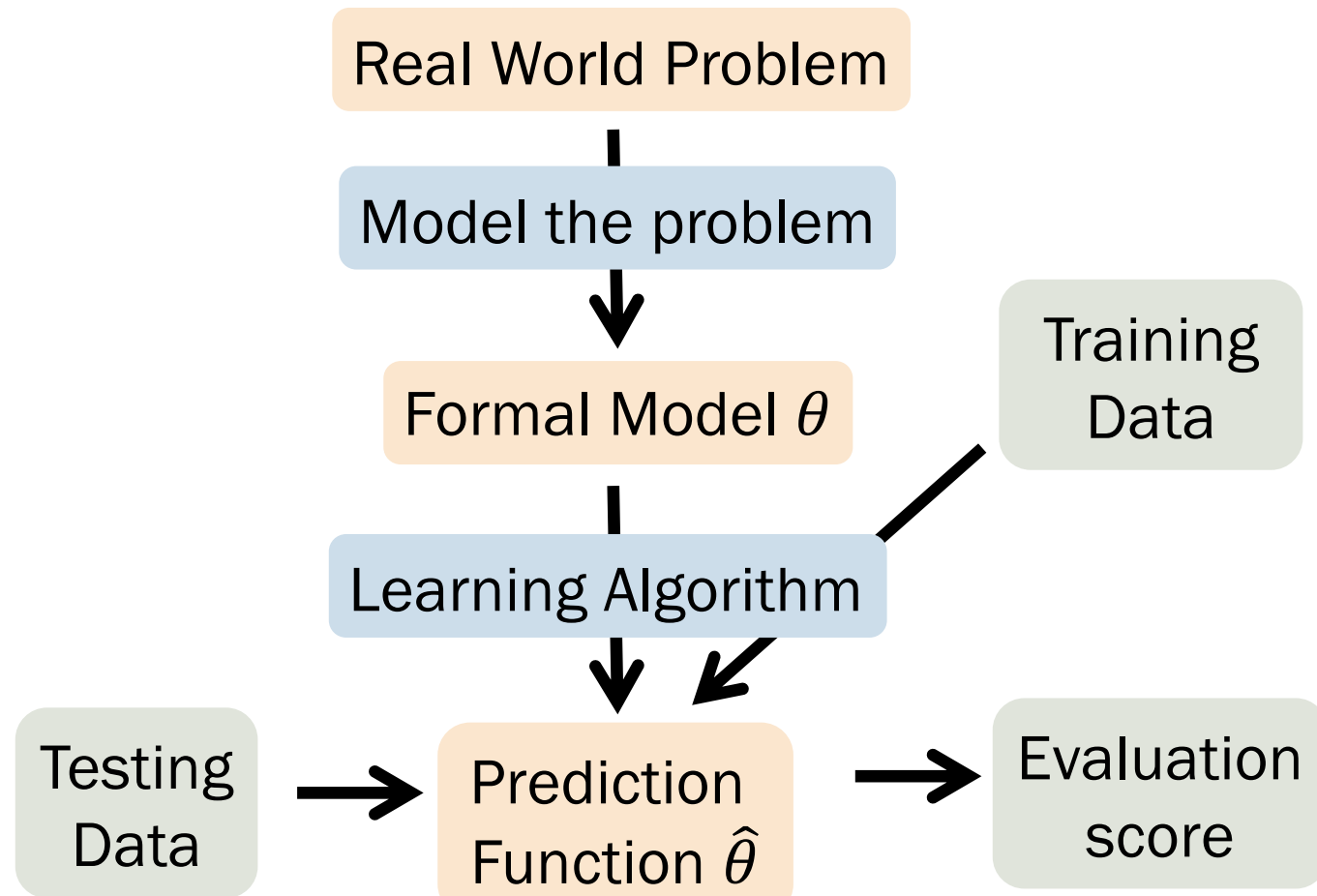


Task: Identify the chair

Data: All the chairs ever

Supervised learning: A category of machine learning where you have labeled data for the problem you are solving.

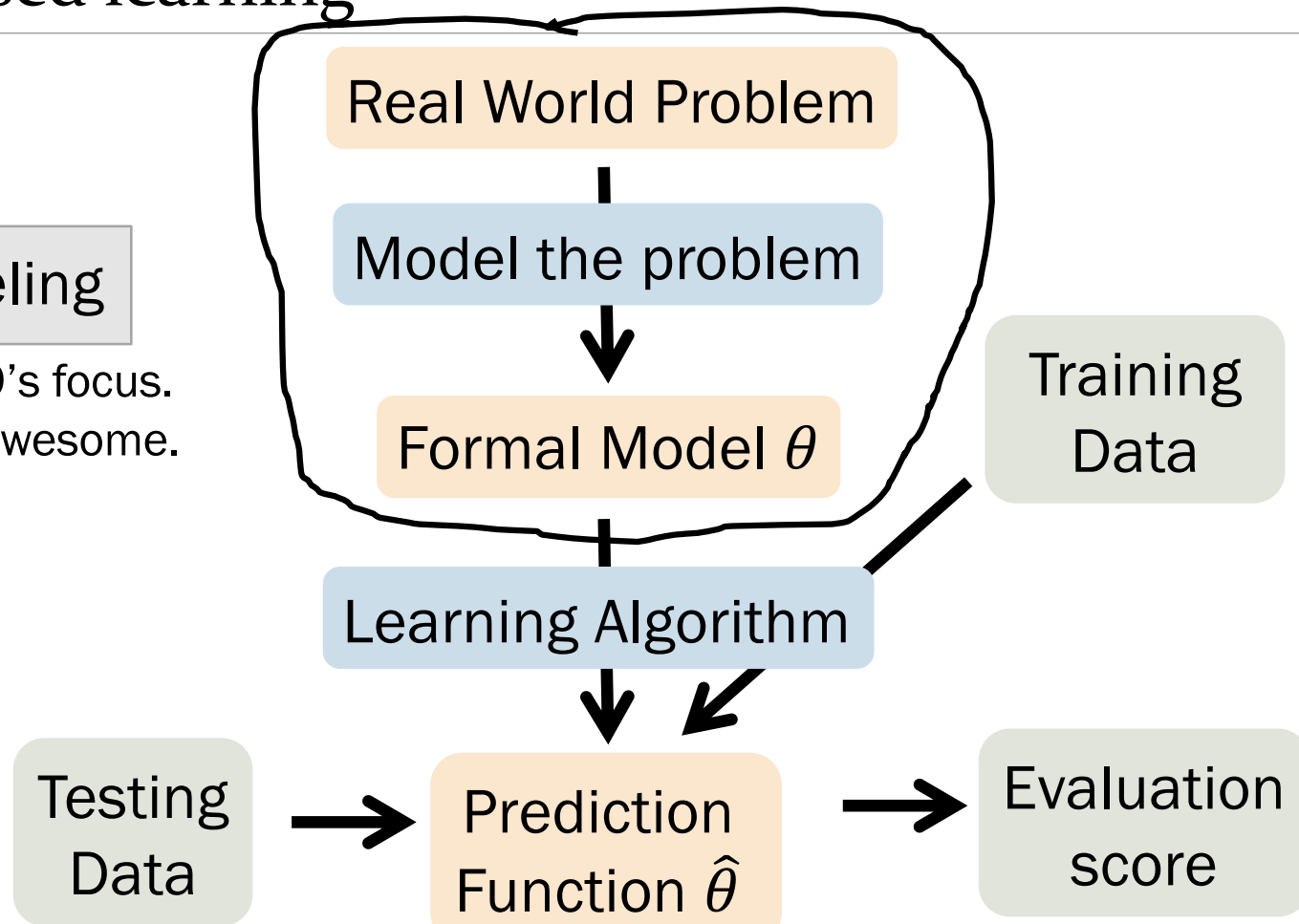
Supervised learning



Supervised learning

Modeling

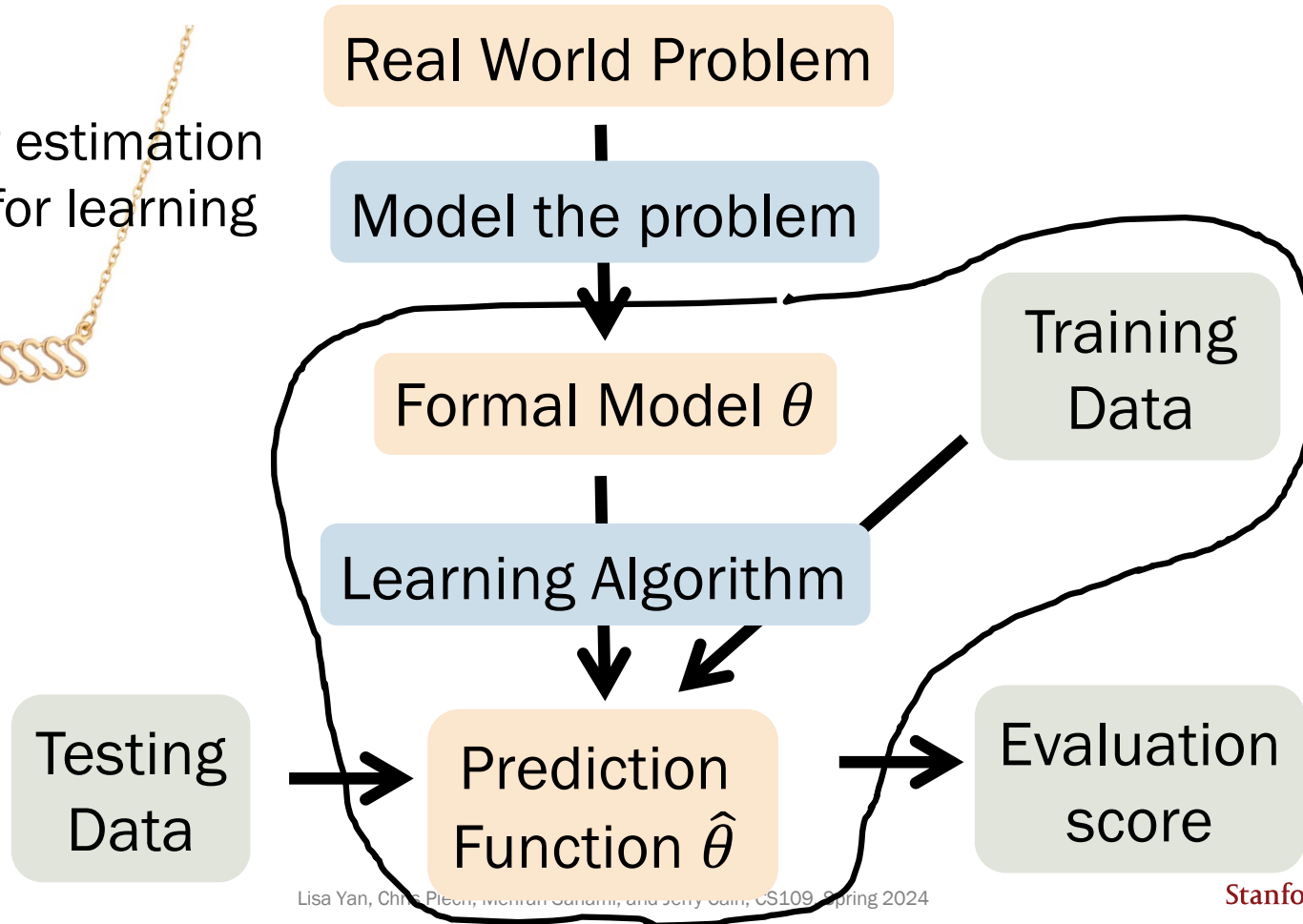
Not CS109's focus.
[CS228](#) is awesome.



Supervised learning

Parameter estimation
is a basis for learning
from data.

Training



Model and dataset

Many different forms of machine learning

- We focus on one specific type: **prediction** from past observations.

Goal

Based on observed $\mathbf{X}$, predict some unknown Y

- **Features**

Vector $\mathbf{X}$ of m observations (new term: **feature vector**)

$$\mathbf{X} = (X_1, X_2, \dots, X_m)$$

- **Output**

Variable Y (also called **class label** if discrete)

Model

$$\hat{Y} = g(\mathbf{X}), \text{ a function on } \mathbf{X}$$

examples:

dental X-rays $\rightarrow$ predict cavities

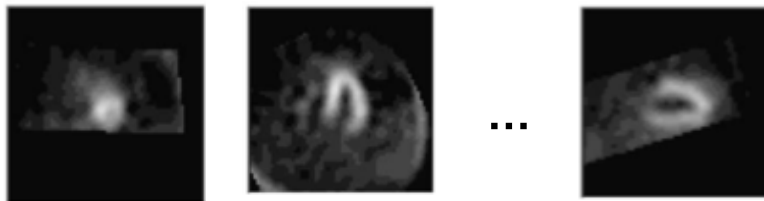
most recently typed characters $\rightarrow$ predict word being typed

temperatures $\rightarrow$ predict tomorrow's weather

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Training data

$$X = (X_1, X_2, X_3, \dots, X_{300})$$



Feature 1 Feature 2 ... Feature 300

We assume all input features are Bernoulli random variables.

Patient 1	1	0	...	1	1
Patient 2	1	1	...	0	0
...			⋮		⋮
Patient n	0	0	...	1	1



Output

Training data notation

in bold, because it's a vector!

$$(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})$$

n datapoints, assumed to be iid

m-dimensional observation

single output

i -th datapoint $(\mathbf{x}^{(i)}, y^{(i)})$: *i th datapoint's second feature*

- m features: $\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_m^{(i)})$
- A single output $y^{(i)}$ *output (label) for i th datapoint*
- Independent of all other datapoints

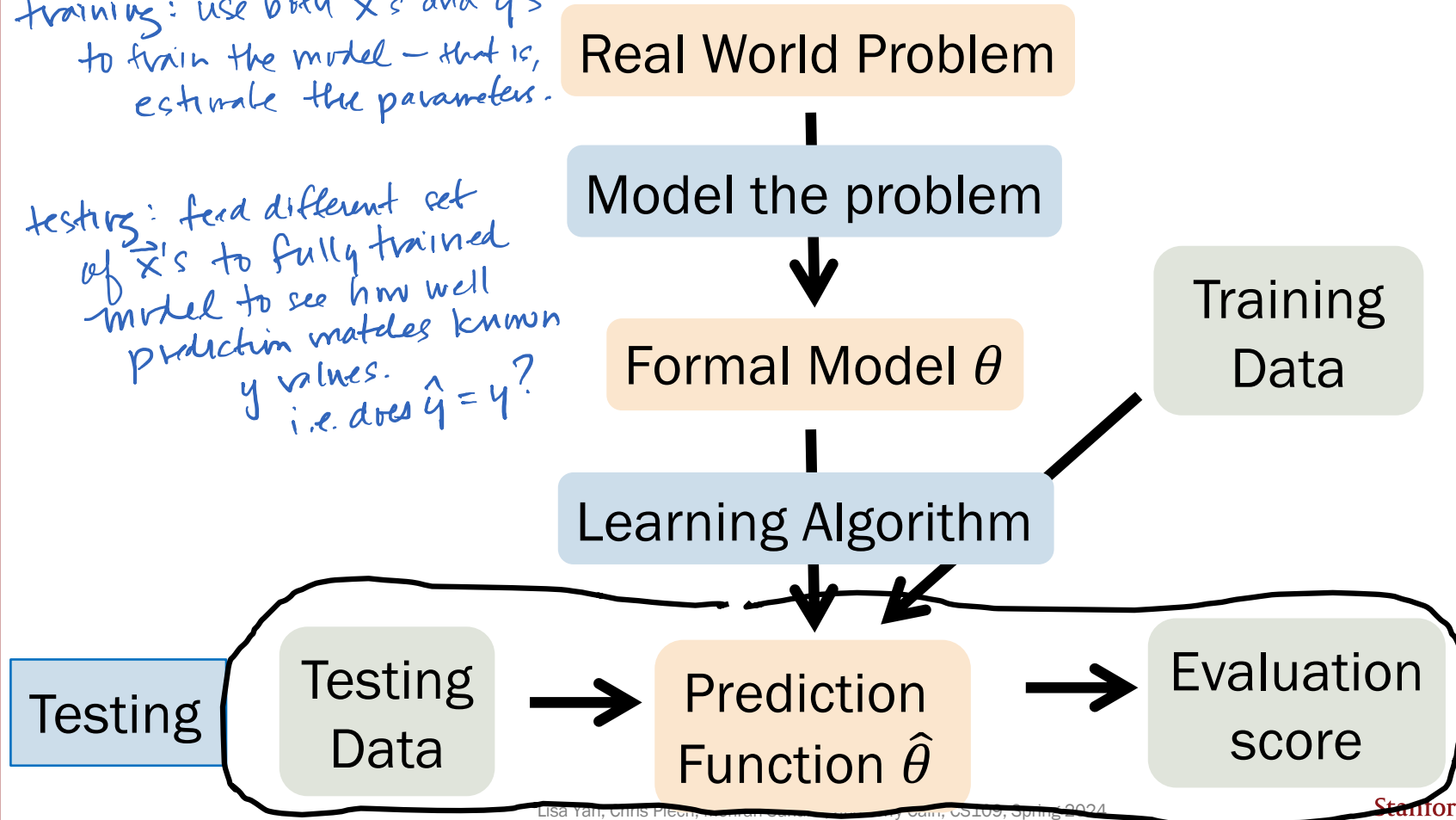
Training Goal:

Use these n datapoints to learn a model $\hat{Y} = g(\mathbf{X})$ that predicts Y

Supervised learning

training: use both $\vec{x}$'s and y 's
to train the model — that is,
estimate the parameters.

testing: feed different set
of $\vec{x}$'s to fully trained
model to see how well
prediction matches known
 y values.
i.e. does $\hat{y} = y$?



Testing data notation

$(\mathbf{x}^{(1)}, y^{(1)}), (\mathbf{x}^{(2)}, y^{(2)}), \dots, (\mathbf{x}^{(n)}, y^{(n)})$
 n_{test} other datapoints, assumed to be iid

i^{th} datapoint $(\mathbf{x}^{(i)}, y^{(i)})$:

- Has the same structure as your training data

Testing Goal:

Leveraging the model $\hat{Y} = g(\mathbf{X})$ that you trained, see how well you can predict Y on known data

Two tasks we will focus on

Many different forms of machine learning

- We focus on the problem of **prediction** based on observations.

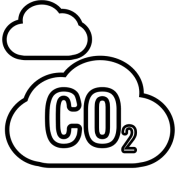
Goal	Based on observed $\mathbf{X}$, predict some unknown Y
• Features	Vector $\mathbf{X}$ of m observations (new term: feature vector) $\mathbf{X} = (X_1, X_2, \dots, X_m)$
• Output	Variable Y (also called class label if discrete)

Model	$\hat{Y} = g(\mathbf{X})$, a function on $\mathbf{X}$
• Regression	prediction when Y is continuous
• Classification	prediction when Y is discrete <i>today's focus</i>

neat! we'll study this a bit next lecture

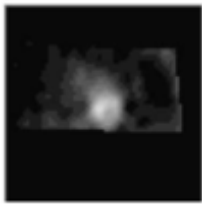
Regression: Predicting real numbers

Training data: $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})$

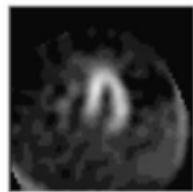
	 CO2 levels	 Sea level	...	 Feature m	 Global Land-Ocean temperature
					Output
Year 1	338.8	0	...	1	0.26
Year 2	340.0	1	...	0	0.32
...			⋮		⋮
Year n	340.76	0	...	1	0.14

Classification: Predicting class labels

$$\mathbf{X} = (X_1, X_2, X_3, \dots, X_{300})$$

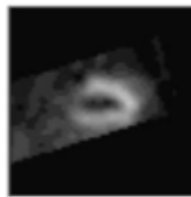


Feature 1



Feature 2

...



Feature 300

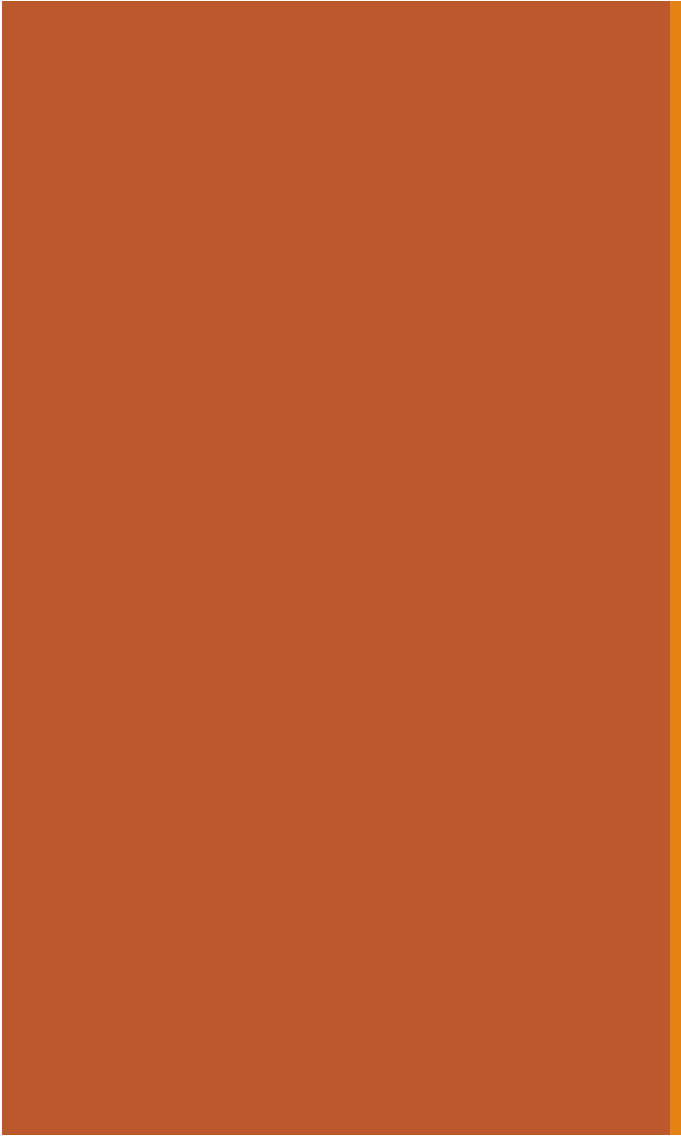


Output

*outputs (aka labels)
aren't always binary,
but they are in
today's examples.*

Patient 1	1	0	...	1
Patient 2	1	1	...	0
...			⋮	
Patient n	0	0	...	1

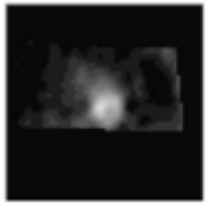
1
0
⋮
1



Brute Force Bayes

Classification: Healthy hearts

$$\mathbf{X} = (X_1)$$



Feature 1



Output

Patient 1	1	0
Patient 2	1	1
	$\vdots$	$\vdots$
Patient n	0	1

Single feature: Region of Interest (ROI) is healthy (1) or unhealthy (0)

How can we predict whether heart is healthy (1) or not (0)?

The following strategy is **not used in practice** but helps us understand how to approach classification.

Classification: Brute Force Bayes

$$\hat{Y} = g(\mathbf{X})$$

Our prediction for Y
is a function of $\mathbf{X}$

$$= \arg \max_{y=\{0,1\}} P(Y | \mathbf{X})$$

Proposed model: Choose the Y that is
more or most likely given $\mathbf{X}$

$$= \arg \max_{y=\{0,1\}} \frac{P(\mathbf{X}|Y)P(Y)}{P(\mathbf{X})}$$

(Bayes' Theorem)

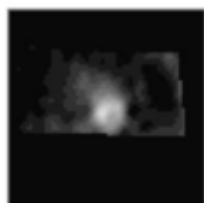
$$= \arg \max_{y=\{0,1\}} P(\mathbf{X}|Y)P(Y)$$

($1/P(\mathbf{X})$ is constant wrt y)

If we estimate $P(\mathbf{X}|Y)$ and $P(Y)$, we can classify data points.

Training: Estimate parameters

$$\mathbf{X} = (X_1)$$



Feature 1



Output

Patient 1 1
 Patient 2 1
 ⋮
 Patient n 0

0
 1
 ⋮
 1

Conditional
 probability
 tables $\hat{P}(\mathbf{X}|Y)$

Marginal
 probability
 table $\hat{P}(Y)$

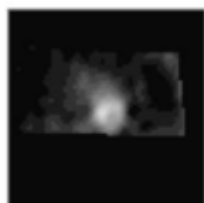
$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(\mathbf{X}|Y) \hat{P}(Y)$$

	$\hat{P}(\mathbf{X} Y = 0)$	$\hat{P}(\mathbf{X} Y = 1)$
$X_1 = 0$	θ_1	θ_3
$X_1 = 1$	$\theta_2 = 1 - \theta_1$	$\theta_4 = 1 - \theta_3$
$\hat{P}(Y)$		
$Y = 0$	θ_5	
$Y = 1$	$\theta_6 = 1 - \theta_5$	

Training
 Goal:

Use n datapoints to learn
 $2 \cdot 2 + 2 = 6$ parameters.

Training: Estimate parameters $\hat{P}(X|Y)$



	$\hat{P}(X Y = 0)$	$\hat{P}(X Y = 1)$
$X_1 = 0$	θ_1	θ_3
$X_1 = 1$	$\theta_2 = 1 - \theta_1$	$\theta_4 = 1 - \theta_3$

	Count:	# datapoints
Pa	$X_1 = 0, Y = 0:$	4
	$X_1 = 1, Y = 0:$	6
Pa	$X_1 = 0, Y = 1:$	0
	$X_1 = 1, Y = 1:$	100
	Total:	110

10 datapoints in $Y=0$ world
100 data points in $Y=1$ world

$X|Y = 0$ and $X|Y = 1$
are each multinomials with 2 outcomes!
(multinomial in general, though binomial when feature vector is of length 1).

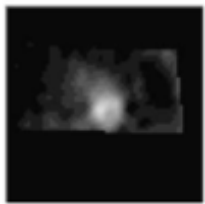
Patient n 0

1

this amounts to brute force counting $\Rightarrow$ are multinomials

Use MLE or Laplace (MAP) estimate $\hat{P}(X|Y)$ and $\hat{P}(Y)$ as parameters.

Training: MLE estimates, $\hat{P}(\mathbf{X}|Y)$



Count:	# datapoints
$X_1 = 0, Y = 0$:	4
$X_1 = 1, Y = 0$:	6
$X_1 = 0, Y = 1$:	0
$X_1 = 1, Y = 1$:	100
Total:	110

MLE

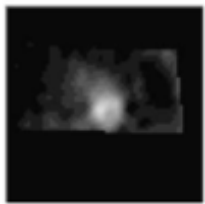
	$\hat{P}(\mathbf{X} Y = 0)$	$\hat{P}(\mathbf{X} Y = 1)$
$X_1 = 0$	0.4 = $\frac{4}{4+6}$	0.0 = $\frac{0}{0+100}$
$X_1 = 1$	0.6 = $\frac{6}{4+6}$	1.0 = $\frac{100}{0+100}$

MLE of $\hat{P}(X_1 = x|Y = y) = \frac{\#(X_1 = x, Y = y)}{\#(Y = y)}$
Just count!

Patient n 0

1

Training: Laplace (MAP) estimates, $\hat{P}(X|Y)$



Count:	# datapoints
$X_1 = 0, Y = 0$:	4
$X_1 = 1, Y = 0$:	6
$X_1 = 0, Y = 1$:	0
$X_1 = 1, Y = 1$:	100
Total:	110

Pa
Pa

Patient n 0

1

MLE

MLE of $\hat{P}(X_1 = x|Y = y) = \frac{\#(X_1 = x, Y = y)}{\#(Y = y)}$
Just count!

MAP

example

$$P(X_1 = 1 | Y = 0) = \frac{\#(X_1 = 1, Y = 0) + 1}{\#(Y = 0) + 1}$$

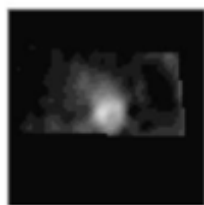
$$= \frac{6 + 1}{4 + 2} = \frac{7}{12}$$

Laplace of $\hat{P}(X_1 = x|Y = y) = ?$

Just count + add imaginary trials!



Training: Laplace (MAP) estimates, $\hat{P}(\mathbf{X}|\mathbf{Y})$



Count:	# datapoints
$X_1 = 0, Y = 0$:	4
$X_1 = 1, Y = 0$:	6
$X_1 = 0, Y = 1$:	0
$X_1 = 1, Y = 1$:	100
Total:	110

MLE

$$\text{MLE of } \hat{P}(X_1 = x|Y = y) = \frac{\#(X_1 = x, Y = y)}{\#(Y = y)}$$

Just count!

	$\hat{P}(\mathbf{X} \mathbf{Y} = 0)$	$\hat{P}(\mathbf{X} \mathbf{Y} = 1)$
$X_1 = 0$	0.4	0.0
$X_1 = 1$	0.6	1.0

MAP

	$\hat{P}(\mathbf{X} \mathbf{Y} = 0)$	$\hat{P}(\mathbf{X} \mathbf{Y} = 1)$
$X_1 = 0$	0.42 $\frac{5}{12}$	0.01 $= \frac{1}{102}$
$X_1 = 1$	0.58 $= \frac{7}{12}$	0.99 $\frac{101}{102}$

$$\text{Laplace of } \hat{P}(X_1 = x|Y = y) = \frac{\#(X_1 = x, Y = y) + 1}{\#(Y = y) + 2}$$

Just count + add imaginary trials!

Testing

*we go with MAP with smoothing for input features
↓ more often than MLE to avoid zero probabilities*

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(X|Y) \hat{P}(Y)$$

MAP with smoothing shouldn't be necessary unless you have zero instances of some outputs.

(MAP)	$\hat{P}(X Y = 0)$	$\hat{P}(X Y = 1)$
$X_1 = 0$	0.42	0.01
$X_1 = 1$	0.58	0.99

(MLE)	$\hat{P}(Y)$
$Y = 0$	0.09 = $\frac{10}{110}$
$Y = 1$	0.91 = $\frac{100}{110}$

New patient has a healthy ROI ($X_1 = 1$). What is your prediction, $\hat{Y}$?

$$\hat{P}(X_1 = 1|Y = 0) \hat{P}(Y = 0) = 0.58 \cdot 0.09 \approx 0.052$$

$$\hat{P}(X_1 = 1|Y = 1) \hat{P}(Y = 1) = 0.99 \cdot 0.91 \approx 0.901$$

- A. $0.052 < 0.5 \Rightarrow \hat{Y} = 1$
- B. $0.901 > 0.5 \Rightarrow \hat{Y} = 1$
- C. $0.052 < 0.901 \Rightarrow \hat{Y} = 1$**

we go with $\hat{Y}=1$ because it comes with a higher probability that's just what binary argmax is!

Brute Force Bayes classifier

work on previous slides dealt
with $m=1$ feature, but
everything generalizes to $m>1$ features.

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(\mathbf{X}|Y) \hat{P}(Y)$$

$(\hat{P}(Y))$ is an estimate of $P(Y)$,
 $\hat{P}(\mathbf{X}|Y)$ is an estimate of $P(\mathbf{X}|Y)$

Training

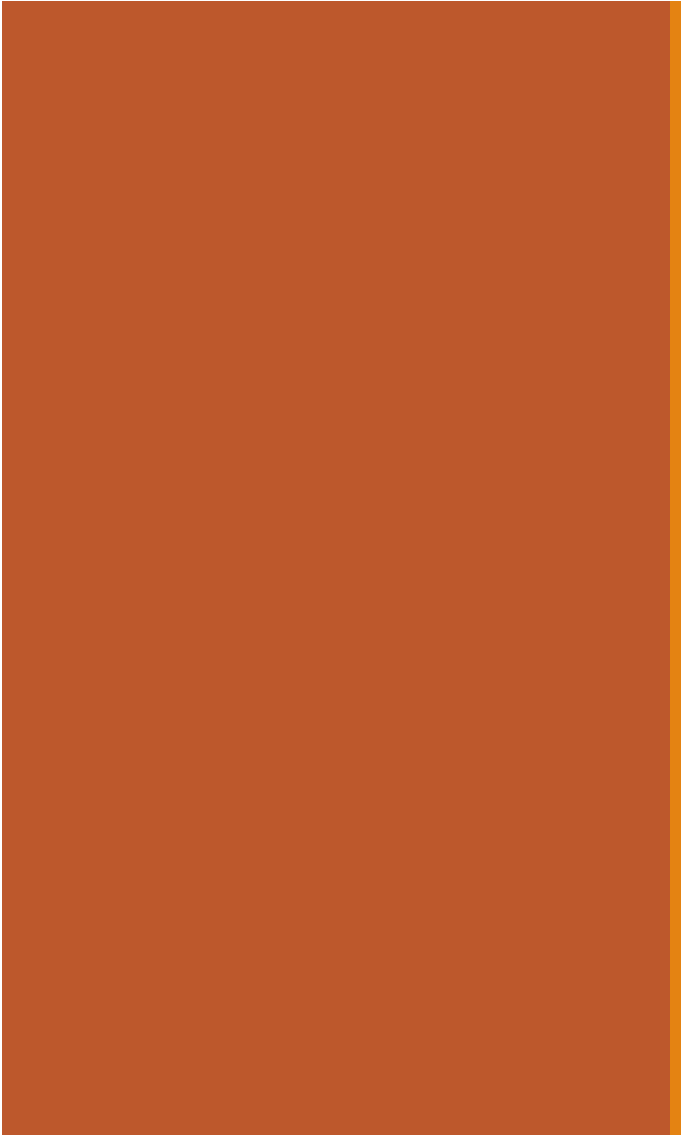
Estimate these
probabilities—i.e.,
learn these parameters
using MLE or Laplace (MAP)

$$\begin{aligned} &\hat{P}(X_1, X_2, \dots, X_m | Y = 1) \\ &\hat{P}(X_1, X_2, \dots, X_m | Y = 0) \\ &\hat{P}(Y = 1) \quad \hat{P}(Y = 0) \end{aligned}$$

Testing

Given an observation $\mathbf{X} = (X_1, X_2, \dots, X_m)$, predict

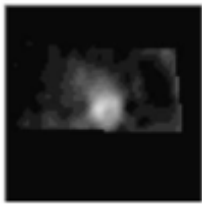
$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\hat{P}(X_1, X_2, \dots, X_m | Y) \hat{P}(Y) \right)$$



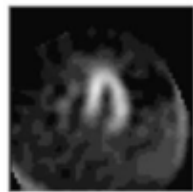
Naïve Bayes

Brute Force Bayes: $m = 300$ (# features)

$$\mathbf{X} = (X_1, X_2, X_3, \dots, X_{300})$$

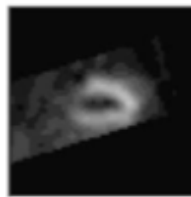


Feature 1



Feature 2

...



Feature 300



Output

Patient 1	1	0	...	1	1
Patient 2	1	1	...	0	0
...			⋮		⋮
Patient n	0	0	...	1	1

This won't be
too bad, right?

Brute Force Bayes: $m = 300$ (# features)

$$\mathbf{X} = (X_1, X_2, X_3, \dots, X_{300})$$



Count:

datapoints

$X_1 = 0, X_2 = 0, \dots, X_{299} = 0, X_{300} = 0, Y = 0:$ 0

$X_1 = 0, X_2 = 0, \dots, X_{299} = 0, X_{300} = 1, Y = 0:$ 0

$X_1 = 0, X_2 = 0, \dots, X_{299} = 1, X_{300} = 0, Y = 0:$ 1

Pat ...

$X_1 = 0, X_2 = 0, \dots, X_{299} = 0, X_{300} = 0, Y = 1:$ 2

Pat

$X_1 = 0, X_2 = 0, \dots, X_{299} = 0, X_{300} = 1, Y = 1:$ 1

$X_1 = 0, X_2 = 0, \dots, X_{299} = 1, X_{300} = 0, Y = 1:$ 1

Patient n

0

0

...

1

1

This won't be too bad, right?

Brute Force Bayes: $m = 300$ (# features)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(Y | \mathbf{X})$$

$$= \arg \max_{y=\{0,1\}} \frac{\hat{P}(\mathbf{X}|Y)\hat{P}(Y)}{\hat{P}(\mathbf{X})}$$

$$= \arg \max_{y=\{0,1\}} \underbrace{\hat{P}(\mathbf{X}|Y)\hat{P}(Y)}$$

Learn parameters
through MLE or MAP

- $\hat{P}(Y = 1 | \mathbf{x})$: estimated probability a heart is healthy given $\mathbf{x}$
- $\mathbf{X} = (X_1, X_2, \dots, X_{300})$: whether 300 regions of interest (ROI) are healthy (1) or unhealthy (0)

How many parameters do we have to learn?

- | | $\hat{P}(\mathbf{X} Y)$ | $\hat{P}(Y)$ | |
|----|-------------------------|--------------|------------------|
| A. | $2 \cdot 2$ | $+ 2$ | $= 6$ |
| B. | $2 \cdot 300$ | $+ 2$ | $= 602$ |
| C. | $2 \cdot 2^{300}$ | $+ 2$ | $= \text{a lot}$ |



Brute Force Bayes: $m = 300$ (# features)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(Y | \mathbf{X})$$

$$= \arg \max_{y=\{0,1\}} \frac{\hat{P}(\mathbf{X}|Y)\hat{P}(Y)}{\hat{P}(\mathbf{X})}$$

$$= \arg \max_{y=\{0,1\}} \underbrace{\hat{P}(\mathbf{X}|Y)\hat{P}(Y)}$$

Learn parameters
through MLE or MAP

This approach requires you to learn
 $O(2^m)$ parameters.

- $\hat{P}(Y = 1 | \mathbf{x})$: estimated probability a heart is healthy given $\mathbf{x}$
- $\mathbf{X} = (X_1, X_2, \dots, X_{300})$: whether 300 regions of interest (ROI) are healthy (1) or unhealthy (0)

How many parameters do we have to learn?

$$\hat{P}(\mathbf{X}|Y) \quad \hat{P}(Y)$$

A. $2 \cdot 2 + 2 = 6$

B. $2 \cdot 300 + 2 = 602$

C. $2 \cdot 2^{300} + 2 = \text{a lot}$

each of 300 features can be either 0 or 1 $\Rightarrow 2^{300}$,
for each of $Y=0$ and $Y=1$.
plus we need estimates for $\hat{P}(Y=0)$ and $\hat{P}(Y=1)$

The problem with our current classifier

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(Y | X)$$

Choose the Y that is most likely given X

$$= \arg \max_{y=\{0,1\}} \frac{\hat{P}(X|Y)\hat{P}(Y)}{\hat{P}(X)}$$

(Bayes' Theorem)

$$= \arg \max_{y=\{0,1\}} \hat{P}(X|Y)\hat{P}(Y)$$

($1/P(X)$ is constant w.r.t. y)



$$\hat{P}(X_1, X_2, \dots, X_m | Y)$$

Estimating this joint conditional distribution is intractable. *for large m*

What if we could make a simplifying assumption—even if incredibly naïve—to make our parameter estimation effort computationally tractable?

The Naïve Bayes assumption

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(Y | X)$$

$$= \arg \max_{y=\{0,1\}} \frac{\hat{P}(X|Y)\hat{P}(Y)}{\hat{P}(X)}$$

$$= \arg \max_{y=\{0,1\}} \hat{P}(X|Y)\hat{P}(Y)$$

$$= \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Assumption:

$X_1, \dots, X_m$ are all **conditionally independent** given Y .

$$\hat{P}(X_1, X_2, X_3, \dots, X_m | Y) \\ = \prod_{i=1}^m \hat{P}(X_i | Y)$$

X_i are very often only mildly conditionally dependent given Y

Naïve Bayes Assumption: by making this assumption, the number of parameters to compute becomes linear in m instead of exponential.

Naïve Bayes Classifier

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training

What is the Big-O of # of parameters we need to learn?

- A. $O(m)$
- B. $O(2^m)$
- C. other



Naïve Bayes Classifier

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training

for $j = 1, \dots, m$:

$$\hat{P}(X_j = 1|Y = 0),$$

$$\hat{P}(X_j = 1|Y = 1)$$

Use MLE or
Laplace (MAP)

$$\hat{P}(Y = 1)$$

$$\hat{P}(Y = 0) = 1 - \hat{P}(Y = 1)$$

$$\hat{P}(X_j = 0|Y = 0) = 1 - \hat{P}(X_j = 1|Y = 0)$$

$$\hat{P}(X_j = 0|Y = 1) = 1 - \hat{P}(X_j = 1|Y = 1)$$

↑

#parameters to be
computed is $4m + 2 = \Theta(m)$

Testing

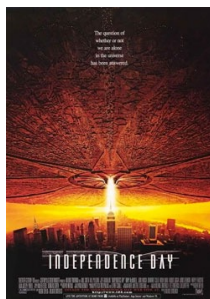
$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\log \hat{P}(Y) + \sum_{j=1}^m \log \hat{P}(X_j|Y) \right)$$

NETFLIX

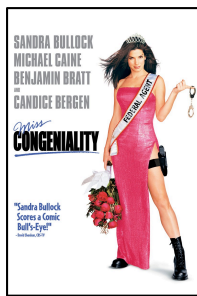
and Learn

Classification terminology check

Training data: $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(n)}, y^{(n)})$

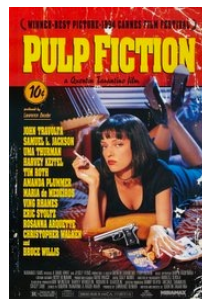


Movie 1



Movie 2

...



Movie m



Output

User 1	1.	1	0	...	1	2.	1
User 2	3.	1	1	...	0		0
...				⋮			⋮
User n		0	4.	0	...	1	1

- A. $x^{(i)}$
- B. $y^{(i)}$
- C. $(x^{(i)}, y^{(i)})$
- D. $x_j^{(i)}$

1: like movie
0: dislike movie

- 1. $\vec{x}^{(1)} \Rightarrow A.$
- 2. $y^{(1)} \Rightarrow B$
- 3. $(\vec{x}^{(2)}, y^{(2)}) \Rightarrow C$
- 4. $x_2^{(m)} \Rightarrow D$



Predicting user TV preferences

Will a user like the Pokémon TV series?

Observe indicator variables $\mathbf{X} = (X_1, X_2)$:



$X_1 = 1$:
"likes Star Wars"



$X_2 = 1$:
"likes Harry Potter"

Output Y indicator:



$Y = 1$:
"likes Pokémon"

Predict $\hat{Y} = \arg \max_{y \in \{0,1\}} \hat{P}(Y | \mathbf{X})$

given X_1 and X_2 , do
we see $Y=0$ or $Y=1$
more often?

Predicting user TV preferences

Which probabilities do you need to estimate?
How many are there?

- Brute Force Bayes
(strawman, without NB assumption)

10

- Naïve Bayes

10

for $m=2$,
numbers
end up being the
same. but
exponential and
is linear.

10

10

During training, how to estimate the prob
 $\hat{P}(X_1 = 1, X_2 = 1 | Y = 0)$ with MLE? with Laplace?

- Brute Force Bayes

$$\text{MLE: } \frac{\#(X_1 = 1, X_2 = 1, Y = 0)}{\#(Y = 1)}$$

$$\text{MAP: } \frac{\#(X_1 = 1, X_2 = 1, Y = 0) + 1}{\#(Y = 0) + 4}$$

- Naïve Bayes

$$\text{MAP: } \frac{\#(X_1 = 1, Y = 0) + 1}{\#(Y = 0) + 2} \cdot \frac{\#(X_2 = 1, Y = 0) + 1}{\#(Y = 0) + 2}$$

MLE is MAP without
+1 and +2

$$\hat{Y} = \arg \max_{y \in \{0,1\}} \hat{P}(X|Y) \hat{P}(Y)$$

Naïve Bayes Assumption $P(X|Y) = \prod_{j=1}^m P(X_j|Y)$

need $\hat{P}(Y=0)$
 $\hat{P}(Y=1)$

$$\hat{P}(X_1=0, X_2=0 | Y=0)$$

$$\hat{P}(X_1=0, X_2=1 | Y=0)$$

$$\hat{P}(X_1=1, X_2=0 | Y=0)$$

$$\hat{P}(X_1=1, X_2=1 | Y=0)$$

total # is $2 \cdot 2^2 + 2 = 10$

and need
similar
for $Y=1$

need $\hat{P}(Y=0)$
 $\hat{P}(Y=1)$

$$\hat{P}(X_1=0 | Y=0)$$

$$\hat{P}(X_1=1 | Y=0)$$

$$\hat{P}(X_2=0 | Y=0)$$

$$\hat{P}(X_2=1 | Y=0)$$

and need
similar
for $Y=1$

total = $2 \cdot 4 + 2 = 10$



Ex 1. Naïve Bayes Classifier (**MLE**)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training

$\forall i: \hat{P}(X_j = 1|Y = 0), \hat{P}(X_j = 0|Y = 0), \hat{P}(X_j = 1|Y = 1), \hat{P}(X_j = 0|Y = 1), \hat{P}(Y = 1), \hat{P}(Y = 0)$ Use **MLE** or Laplace (MAP)

Testing

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training: Naïve Bayes for TV shows (MLE)

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	X_1		$Y \backslash X_2$	X_2	
	0	1		0	1
0	3	10	0	5	8
1	4	13	1	7	10

Training data counts

1. How many datapoints (n) are in our training data?
2. Compute MLE estimates for $\hat{P}(X_1|Y)$:

$Y \backslash X_1$	X_1	
	0	1
0	$\hat{P}(X_1 = 0 Y = 0)$	$\hat{P}(X_1 = 1 Y = 0)$
1	$\hat{P}(X_1 = 0 Y = 1)$	$\hat{P}(X_1 = 1 Y = 1)$



Training: Naïve Bayes for TV shows (MLE)

Observe indicator vars. $X = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	0	1	$Y \backslash X_2$	0	1
	3	10		5	8
0	3	10	0	5	8
1	4	13	1	7	10

Training data counts

13 datapoints for $Y=0$

17 datapoints for $Y=1$

$n = 13 + 17 = 30$

1. How many datapoints (n) are in our training data? $\Rightarrow 30$

2. Compute MLE estimates for $\hat{P}(X_1|Y)$:

$Y \backslash X_1$	0	1
0	$\hat{P}(X_1=0 Y=0) = \frac{3}{13} \approx 0.23$	$10/13 = 0.77$
1	$\hat{P}(X_1=0 Y=1) = \frac{4}{17} \approx 0.24$	$13/17 = 0.76$

Training: Naïve Bayes for TV shows (MLE)

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	0		$Y \backslash X_2$	0		Y	
	0	1		0	1		
0	3	10	0	5	8	0	13
1	4	13	1	7	10	1	17

Training data counts

$Y \backslash X_1$	0	1
0	0.23	0.77
1	0.24	0.76

(from last slide)

$Y \backslash X_2$	0		Y	
	0	1		
0	$5/13 \approx 0.38$	$8/13 \approx 0.62$	0	$13/30 \approx 0.43$
1	$7/17 \approx 0.41$	$10/17 \approx 0.59$	1	$17/30 \approx 0.57$

Training: Naïve Bayes for TV shows (MLE)

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	0	1
0	0.23	0.77
1	0.24	0.76

$Y \backslash X_2$	0	1
0	0.38	0.62
1	0.41	0.59

Y	
0	0.43
1	0.57

Now that we’ve trained and found parameters,
It’s time to classify new users!

Ex 1. Naïve Bayes Classifier (**MLE**)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training

$\forall i: \hat{P}(X_j = 1|Y = 0), \hat{P}(X_j = 0|Y = 0), \hat{P}(X_j = 1|Y = 1), \hat{P}(X_j = 0|Y = 1), \hat{P}(Y = 1), \hat{P}(Y = 0)$ Use **MLE** or Laplace (MAP)

Testing

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Testing: Naïve Bayes for TV shows (MLE)

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	X_1	
	0	1
0	0.23	0.77
1	0.24	0.76

$Y \backslash X_2$	X_2	
	0	1
0	0.38	0.62
1	0.41	0.59

Y	
0	0.43
1	0.57

Suppose a **new person** “likes Star Wars” ($X_1 = 1$) but “dislikes Harry Potter” ($X_2 = 0$).

Will they like Pokemon? Need to predict Y :

$$\hat{Y} = \arg \max_{y=\{0,1\}} \hat{P}(\mathbf{X}|Y) \hat{P}(Y) = \arg \max_{y=\{0,1\}} \hat{P}(X_1|Y) \hat{P}(X_2|Y) \hat{P}(Y)$$

If $Y = 0$: $\hat{P}(X_1 = 1|Y = 0) \hat{P}(X_2 = 0|Y = 0) \hat{P}(Y = 0) = 0.77 \cdot 0.38 \cdot 0.43 = 0.126$

If $Y = 1$: $\hat{P}(X_1 = 1|Y = 1) \hat{P}(X_2 = 0|Y = 1) \hat{P}(Y = 1) = 0.76 \cdot 0.41 \cdot 0.57 = 0.178$

Since term is greatest when $Y = 1$, predict $\hat{Y} = 1$

Ex 2. Naïve Bayes Classifier (MAP)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

Training

$\forall i: \hat{P}(X_j = 1|Y = 0), \hat{P}(X_j = 0|Y = 0), \hat{P}(X_j = 1|Y = 1), \hat{P}(X_j = 0|Y = 1), \hat{P}(Y = 1), \hat{P}(Y = 0)$ Use MLE or Laplace (MAP)

Testing

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{j=1}^m \hat{P}(X_j|Y) \right) \hat{P}(Y)$$

(note the same as before)

Training: Naïve Bayes for TV shows (MAP)

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$Y \backslash X_1$	X_1		$Y \backslash X_2$	X_2	
	0	1		0	1
0	3	10	0	5	8
1	4	13	1	7	10

Training data counts

$\hat{P}(X_j = x | Y = y)$:

A. $\frac{\#(X_j=x, Y=y)}{\#(Y=y)}$

B. $\frac{\#(X_j=x, Y=y)+1}{\#(Y=y)+2}$

C. $\frac{\#(X_j=x, Y=y)+1}{\#(Y=y)+4}$

D. other

What are our MAP estimates using Laplace smoothing for $\hat{P}(X_j | Y)$?

we're only concerned with an estimate for X_j and Y , ignoring the other X_k



Training: Naïve Bayes for TV shows (MAP)

$$\hat{Y} = \arg \max_{y=\{0,1\}} \left(\prod_{i=1}^m \hat{P}(X_i|Y) \right) \hat{P}(Y)$$

Observe indicator vars. $\mathbf{X} = (X_1, X_2)$:

- X_1 : “likes Star Wars”
- X_2 : “likes Harry Potter”

Predict Y : “likes Pokémon”

$X_1 \backslash Y$	0	1
0	3	10
1	4	13

$X_2 \backslash Y$	0	1
0	5	8
1	7	10

Y	
0	13
1	17

Training data counts

$X_1 \backslash Y$	0	1
0	0.27	0.73
1	0.26	0.74

$X_2 \backslash Y$	0	1
0	0.40	0.60
1	0.42	0.58

In practice:

- We use Laplace for $\hat{P}(X_j|Y)$ in case some events $X_j = x_j$ don't appear
- We **don't** use Laplace for $\hat{P}(Y)$, because all class labels should appear reasonably often